AN INVERTIBLE SHEAF ON A PROPER CURVE IS AMPLE EXACTLY WHEN ITS DEGREE IS POSITIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites eight cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Sign and triviality criteria for invertible sheaves on a proper curve. Stacks Project, Varieties, Lemma lemma-check-invertible-sheaf-trivial (tag 0B40). Turns a nonzero section vanishing at a point into strict positivity of the degree.
- The Riemann-Roch type formula for the Euler characteristic of powers. Stacks Project, Varieties, Lemma lemma-degree-on-proper-curve (tag 0AYV). Gives chi(X, L^n) = n deg(L) + chi(X, O_X), which drives the growth estimate.

Source reference. Original reference 01WN: sources/stacks-chow/tags.txt, source lines 2488–2488. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B40: sources/stacks-chow/tags.txt, source lines 13983–13983. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0AYX: sources/stacks-chow/tags.txt, source lines 13805–13805. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01TF: sources/stacks-chow/tags.txt, source lines 2372–2372. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01PV: sources/stacks-chow/tags.txt, source lines 2244–2244. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0B5U: sources/stacks-chow/tags.txt, source lines 14047–14047. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0B3J: sources/stacks-chow/tags.txt, source lines 13967–13967. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 01XR: sources/stacks-chow/tags.txt, source lines 2530–2530. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X be a proper curve over k. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Then $\mathcal{L}$ is ample if and only if* $\deg(\mathcal{L}) > 0$.

Proof. If $\mathcal{L}$ is ample, then there exists an $n > 0$ and a section $s \in H^0(X, \mathcal{L}^{\otimes n})$ with X_s affine. Since X isn't affine (otherwise by Morphisms, Lemma 01WN X would be finite), we see that s vanishes at some point. Hence $\deg(\mathcal{L}^{\otimes n}) > 0$ by Lemma 0B40. By Lemma 0AYX we conclude that $\deg(\mathcal{L}) = 1/n \deg(\mathcal{L}^{\otimes n}) > 0$.

Assume $\deg(\mathcal{L}) > 0$. Then

$$\dim_k H^0(X, \mathcal{L}^{\otimes n}) \geq \chi(X, \mathcal{L}^n) = n \deg(\mathcal{L}) + \chi(X, \mathcal{O}_X)$$

grows linearly with n. Hence for any finite collection of closed points $x_1, \ldots, x_t$ of X, we can find an n such that $\dim_k H^0(X, \mathcal{L}^{\otimes n}) > \sum \dim_k \kappa(x_i)$. (Recall that by Hilbert Nullstellensatz, the extension fields $\kappa(x_i)/k$ are finite, see for example Morphisms, Lemma 01TF). Hence we can find a nonzero $s \in H^0(X, \mathcal{L}^{\otimes n})$ vanishing in $x_1, \ldots, x_t$. In particular, if we choose $x_1, \ldots, x_t$ such that $X \setminus \{x_1, \ldots, x_t\}$ is affine, then X_s is affine too (for example by Properties, Lemma 01PV although if we choose our finite set such that $\mathcal{L}|_{X \setminus \{x_1, \ldots, x_t\}}$ is trivial, then it is immediate). The conclusion is that we can find an $n > 0$ and a nonzero section $s \in H^0(X, \mathcal{L}^{\otimes n})$ such that X_s is affine.

We will show that for every quasi-coherent sheaf of ideals $\mathcal{I}$ there exists an $m > 0$ such that $H^1(X, \mathcal{I} \otimes \mathcal{L}^{\otimes m})$ is zero. This will finish the proof by Cohomology of Schemes, Lemma 0B5U. To see this we consider the maps

$$\mathcal{I} \xrightarrow{s} \mathcal{I} \otimes \mathcal{L}^{\otimes n} \xrightarrow{s} \mathcal{I} \otimes \mathcal{L}^{\otimes 2n} \xrightarrow{s} \ldots$$

Since $\mathcal{I}$ is torsion free, these maps are injective and isomorphisms over X_s, hence the cokernels have vanishing H^1 (by Cohomology of Schemes, Lemma 0B3J for example). We conclude that the maps of vector spaces

$$H^1(X, \mathcal{I}) \to H^1(X, \mathcal{I} \otimes \mathcal{L}^{\otimes n}) \to H^1(X, \mathcal{I} \otimes \mathcal{L}^{\otimes 2n}) \to \ldots$$

are surjective. On the other hand, the dimension of $H^1(X, \mathcal{I})$ is finite, and every element maps to zero eventually by Cohomology of Schemes, Lemma 01XR. Thus for some $e > 0$ we see that $H^1(X, \mathcal{I} \otimes \mathcal{L}^{\otimes en})$ is zero. This finishes the proof. □